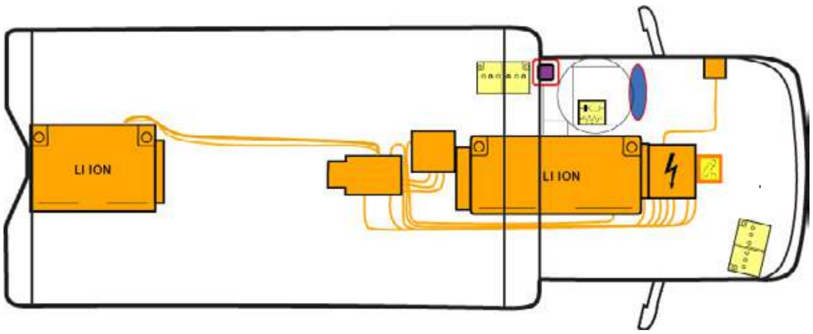
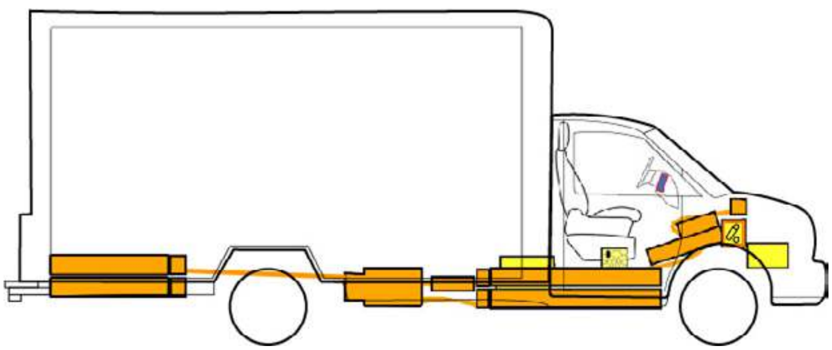
QRG Lightning eMotors Cargo Express 4500
2022-20XX



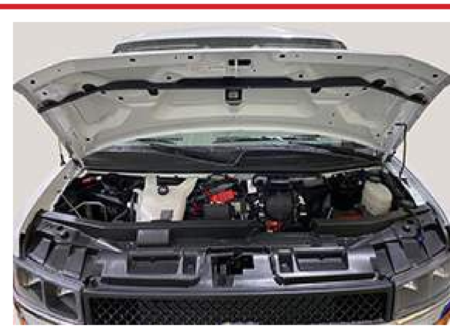
Top View



	Airbag		Battery, low voltage		Seat belt pretensioner		SRS control unit
LI ION	Battery, high voltage		High-voltage power cable		Low voltage disconnect to high voltage		Dangerous high voltage

1. Identification / recognition

The electric motor is silent. Vehicle movement is capable until vehicle is fully shut down.



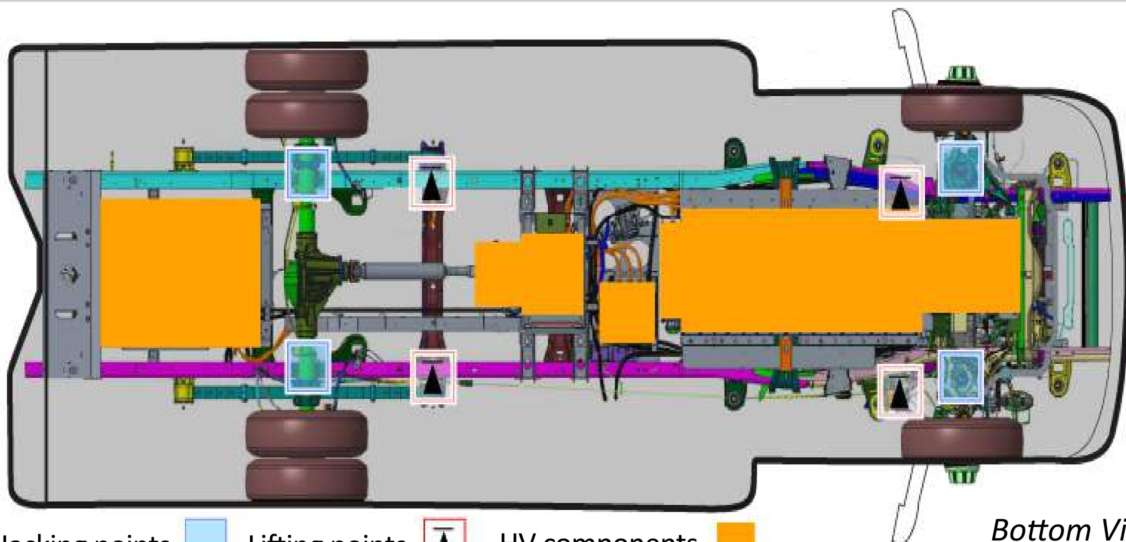
2. Immobilization / stabilization / lifting

Immobilize the vehicle
Apply parking brake
Put vehicle in Park
Turn off vehicle


Parking Brake

Park Selector

Lifting points
(on frame only)



Jacking points | Lifting points | HV components

Bottom View

3. Disable direct hazards / safety regulations

Do not touch, cut, or open high-voltage components and batteries.

Method 1 - Preferred

Step 1: Turn off ignition.

Step 2: Disconnect negative terminals from 12V batteries located on passenger side under the hood and on the driver side behind the cab.

Method 2 - If Method 1 cannot be completed

Step 1: Cut wire in location shown below.

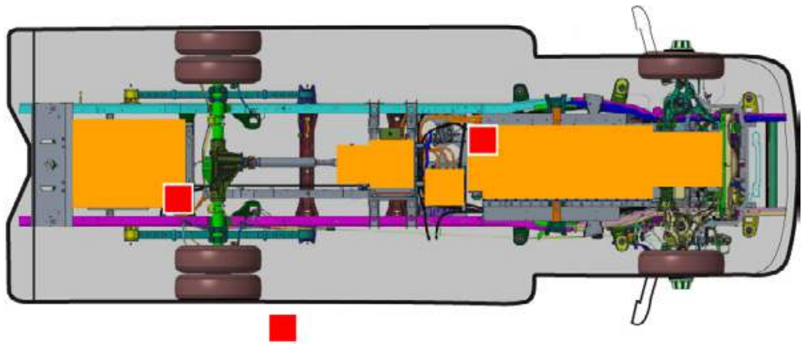
Step 2: Disconnect negative terminals from 12V batteries located on passenger side under the hood and on the driver side behind the cab.


12V locations


Cut wire location

Method 3 - If Methods 1 and 2 cannot be completed

Step 1: If possible, turn off ignition.
Step 2: If possible, disconnect negative terminals from 12V batteries located on passenger side under the hood and on the driver side behind the cab.
Step 3: Remove the stacked Manual Service Disconnects (MSD) from all four batteries.



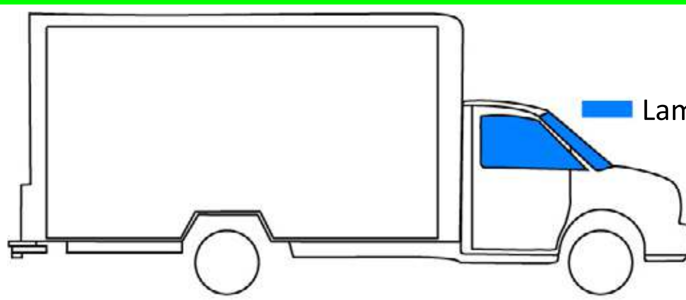
Four (4) MSD locations

Bottom View

MSD

Stacked MSDs

4. Access to occupants

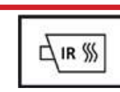


5. Stored energy / liquids / gasses / solids

350V

12V

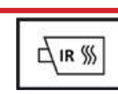
There is an increased risk of a thermal reaction in the high-voltage battery if there is a significant loss of coolant. Monitor temperature of the high-voltage batteries using an infrared camera.



6. In case of fire

Use copious amounts of water directly on high-voltage battery to extinguish fire.

Damaged lithium-ion batteries may re-ignite after the fire is extinguished.
Monitor temperature of the high-voltage batteries using an infrared camera.

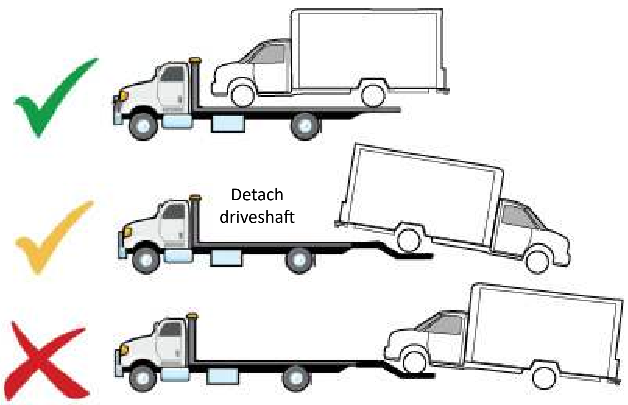


7. In case of submersion

If possible, turn the ignition off in a submerged vehicle to disable the high-voltage system.
No other disabling activities are to be attempted.
Immobilize the vehicle, allow it to dry, and then perform disabling steps in Section 3.

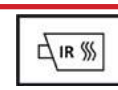
Submerged vehicles pose an increased risk for producing secondary fire events (re-ignition).

8. Towing / transportation / storage



1. Deactivate high-voltage system (see Section 3).
2. Perform an inspection for evidence of active fire or smoldering. Visual or audible clues include gurgling, bubbling, popping, crackling or hissing noises.
3. Confirm high-voltage batteries are not rising above ambient temperature using an infrared camera.
4. Load vehicle for transport.
5. Park vehicle outdoors at least 50 feet from buildings or other vehicles.

Damaged lithium-ion batteries may re-ignite after the fire is extinguished.
Use infrared camera to confirm battery temperature is not rising above ambient before towing.



9. Important additional information

10. Explanation of pictograms used

Flammable

Explosive

Hazardous to human health

Environmental hazard

Acute toxicity

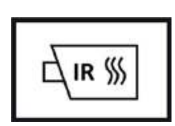
General warning

Use lots of water to extinguish fire

Lithium-ion battery

Corrosive

High-voltage warning



Use infrared camera

Lifting points